CPE134 [Fundamentals of Electrical Circuits]

KIRCHHOFF'S CURRENT AND VOLTAGE LAW

LABORATORY EXERCISE 2

Leader:

Date: _____ Rating: _____

Members:

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NEW MESSAGE

From: **Neha Gupta** (Test Engineering Manager)

Subject: **Demonstrating KCL and KVL**

Good day, _____

We are currently in the process of developing a testing procedure for our product code: "Lotus". As part of this process, we kindly request your team to provide us with **KCL** and **KVL equations** to be incorporated into our test program.

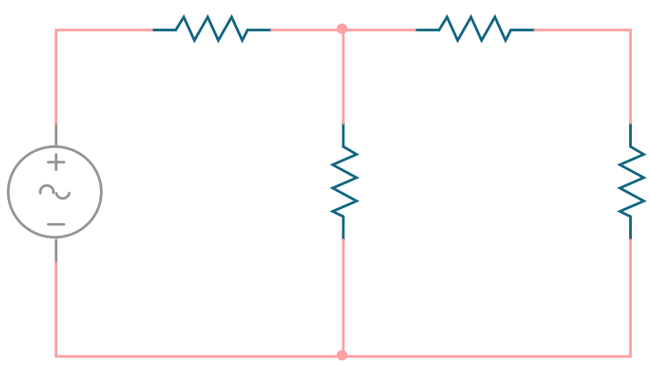
Please take a moment to review the attached files for more detailed information regarding this request.

Best Regards,
Neha

EXERCISE 1 VOLTAGE AND CURRENT MEASUREMENTS

Step 1 **Label** the variables and **set-up** the given circuit.

Circuit 1



Step 2 Measure and record the actual **resistance**,**voltage** and **current** values.

Table 1

Parameters	Value
R_1	
R_2	
R_3	
R_4	
v_1	
v_2	
v_3	
v_4	
i_1	
i_2	
i_3	
i_4	

EXERCISE 2 DEMONSTRATING KIRCHOFF'S CURRENT AND VOLTAGE LAW

Step 1 Formulate the **equation** for Circuit 1 using KCL and KVL.

Step 2 Substitute the **actual values** (from Table 1) to the equation.

Conclusion,